

BMCS2074 · ARTIFICIAL INTELLIGENCE

CONSUMER COMPLAINT URGENCY CLASSIFICATION

Financial complaint triage with classical NLP and DistilBERT

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01 PROBLEM

COMPLAINT TEAMS MUST DISTINGUISH INCONVENIENCE FROM CONTINUING HARM

● Routine issue

“The transfer was briefly held, then released after review.”

A frustrating event, but no continuing material impact.

● Time-sensitive harm

“Unauthorized transactions are still occurring and essential funds are inaccessible.”

The narrative signals ongoing financial and account-access risk.

Product category and sentiment are insufficient. The model must interpret the full narrative.

URGENCY IS DEFINED BY IMMEDIACY AND SEVERITY

● Low

Routine or resolved
Informational questions, minor
inconvenience, or dissatisfaction
without continuing material impact.



● Medium

Significant and unresolved
Priority attention is needed, but the
narrative does not show explicit
immediate severe harm.

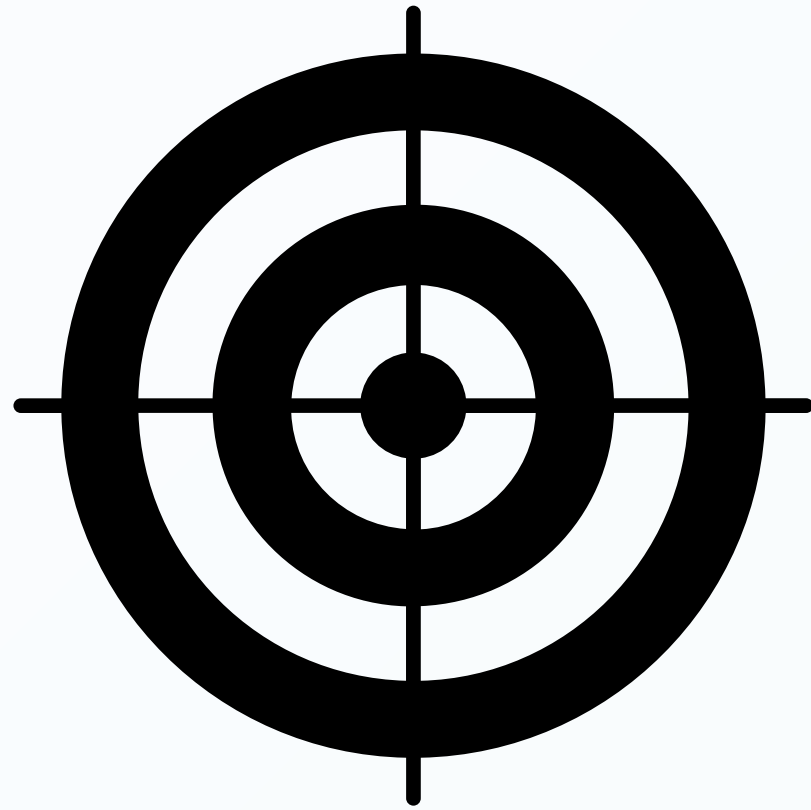


● High

Immediate or continuing risk
Financial, identity, housing, legal,
safety, or essential account-access
harm is active or imminent.



WE EVALUATED MORE THAN HEADLINE ACCURACY



01

How effectively can four NLP methods classify Low, Medium, and High urgency?

		Predicted	
		Positive	Negative
Actual	Positive	TP	FN
	Negative	FP	TN

02

How do they compare on macro-F1, High recall, and serious error directions?



03

What performance, interpretability, confidence, and cost trade-offs matter?

THE MISSING PIECE WAS A CONTROLLED, HARM-BASED URGENCY STUDY

PRIOR EMPHASIS

- Product classification
- Company response prediction
- Topic discovery
- General complaint detection

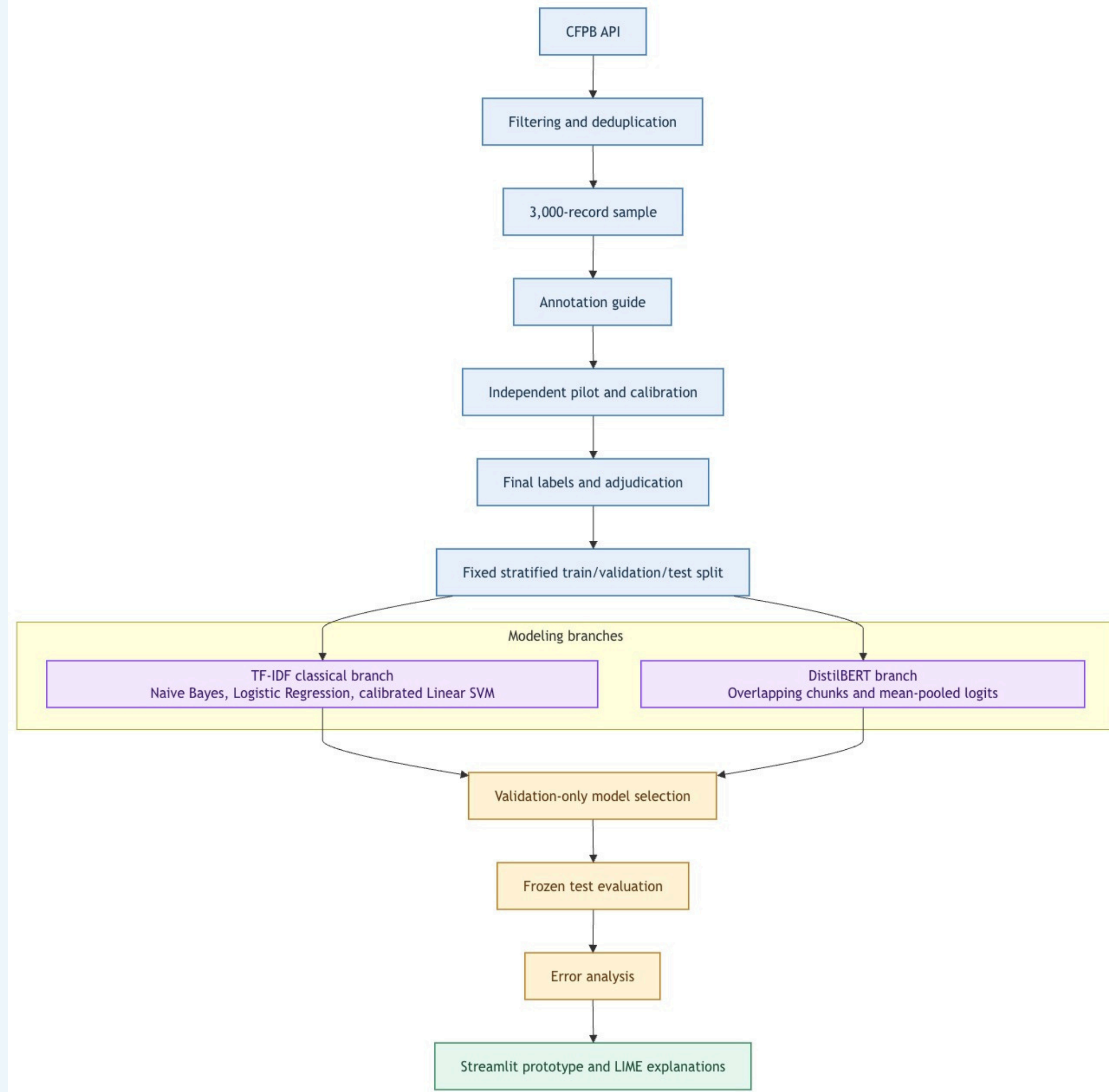
OUR CONTRIBUTION

Three-level urgency based on current harm, evaluated with identical splits and connected to a human-led operations prototype.

02

METHODOLOGY

OUR PIPELINE



WE PRODUCED 3,000 UNIQUE, LABELED NARRATIVES

6,000

Raw complaints retrieved through the CFPB API

-1,445

Duplicate narratives discarded

4,555

Unique narratives in the cleaned pool

3,000

Seeded final sample for annotation and modeling



2022-2025

Complaint receipt window

SHA-256

Processed hashes recorded

20260806

Final sampling seed

REFINING THE GUIDE INCREASED CHANCE-CORRECTED AGREEMENT

INITIAL PILOT • 400 RECORDS

κ **0.4278**

83.25% raw agreement

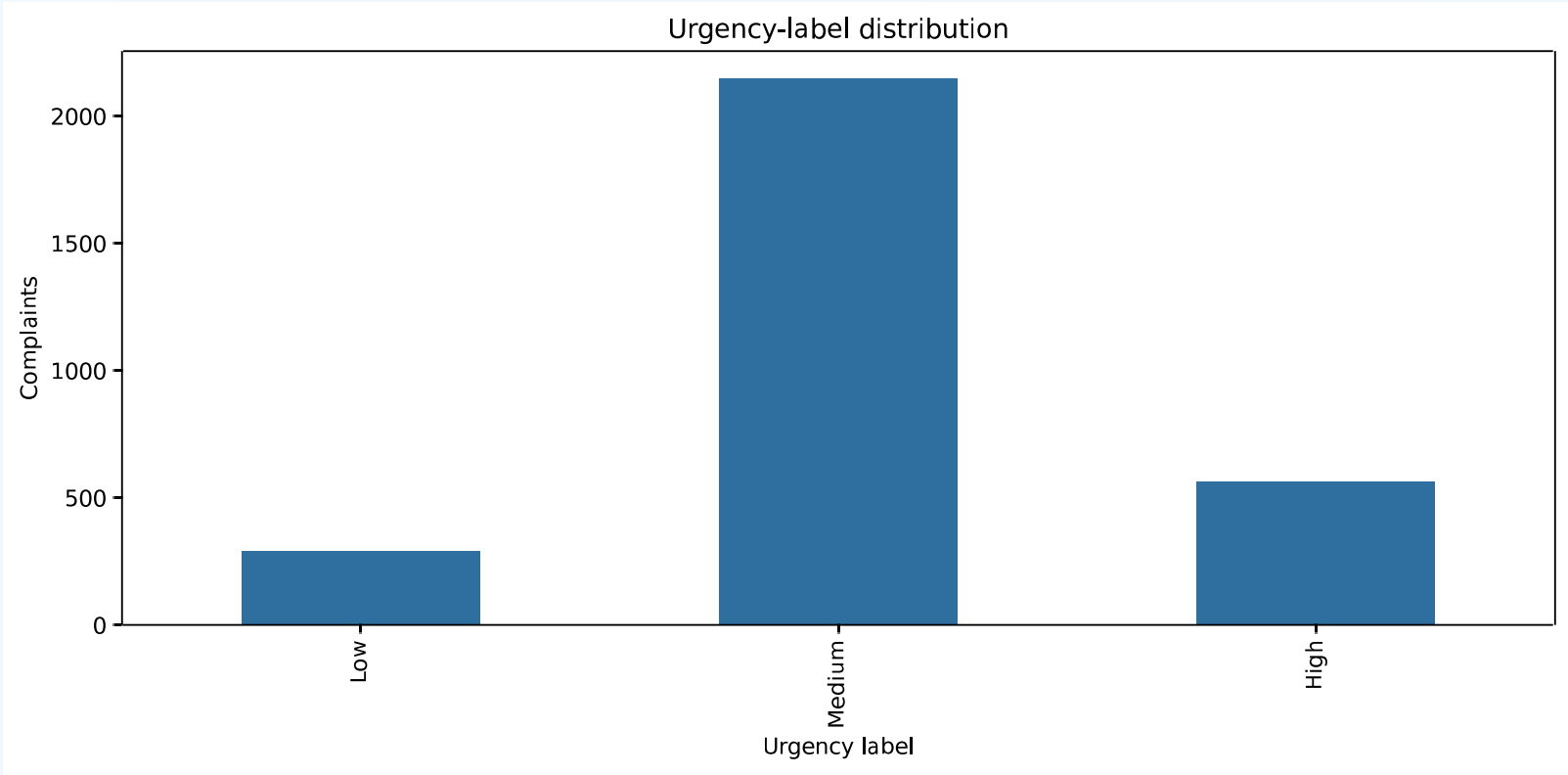
CALIBRATION • 67 RECORDS

κ **0.7485**

83.58% raw agreement

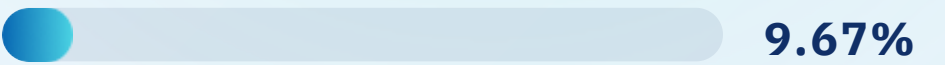
Two independent AI annotators (gpt5.6-terra_low) produced research labels; six uncertain final cases were reviewed and every decision retained provenance.

CLASS IMBALANCE MADE ACCURACY AN UNSAFE PRIMARY METRIC

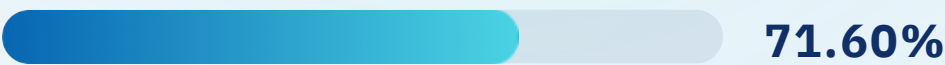


n = 3,000 labeled narratives.

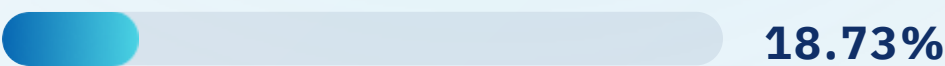
Low



Medium

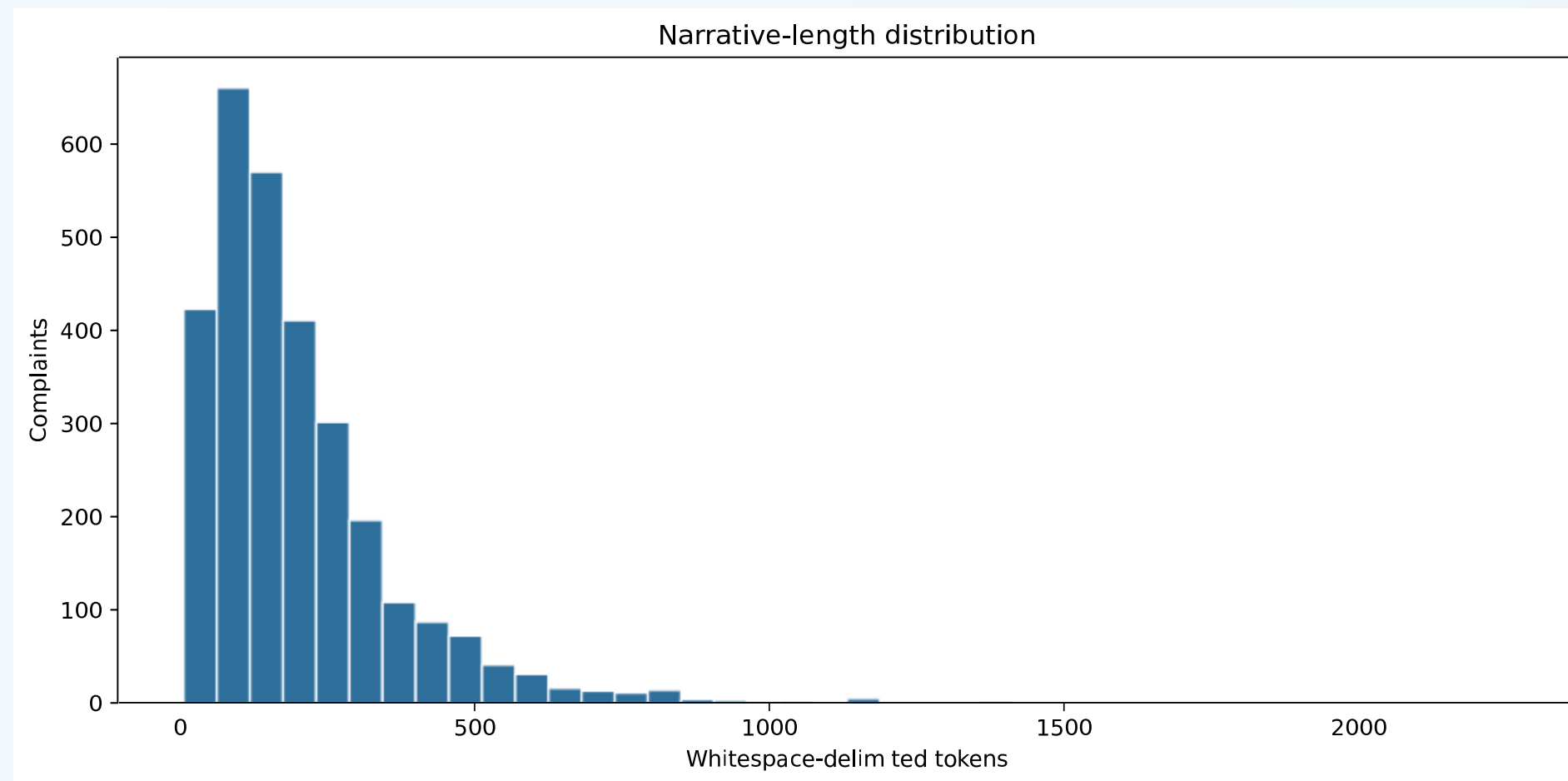


High



Macro-F1 gives each urgency class equal influence.

LONG COMPLAINTS REQUIRED AN EXPLICIT DOCUMENT STRATEGY FOR DISTILLBERT



43.1%

of training narratives exceeded 256 tokens

DistilBERT used overlapping 256-token chunks with a stride of 64, then averaged chunk logits.

CLASSICAL AND CONTEXTUAL MODELS SAW THE SAME NARRATIVES DIFFERENTLY

CLASSICAL PATH

27,493 TF-IDF features

- Lowercase and placeholder normalization
- Unigrams and bigrams
- Training-fitted vocabulary
- Full normalized narrative

TRANSFORMER PATH

Contextual token sequences

- DistilBERT uncased tokenizer
- Minimal text alteration
- Overlapping long-document chunks
- Mean-logit document aggregation

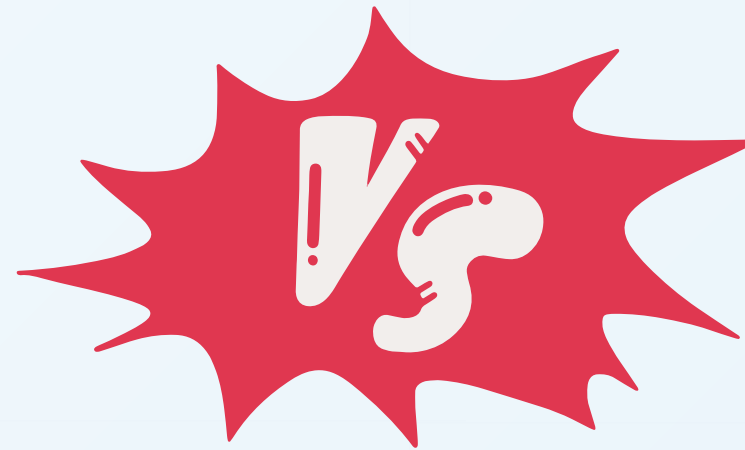
FOUR MODELS TESTED DISTINCT ASSUMPTIONS ABOUT LANGUAGE

Multinomial Naive Bayes

Fast probabilistic baseline for non-negative sparse features.

Logistic Regression

Balanced, regularized, probabilistic linear baseline.



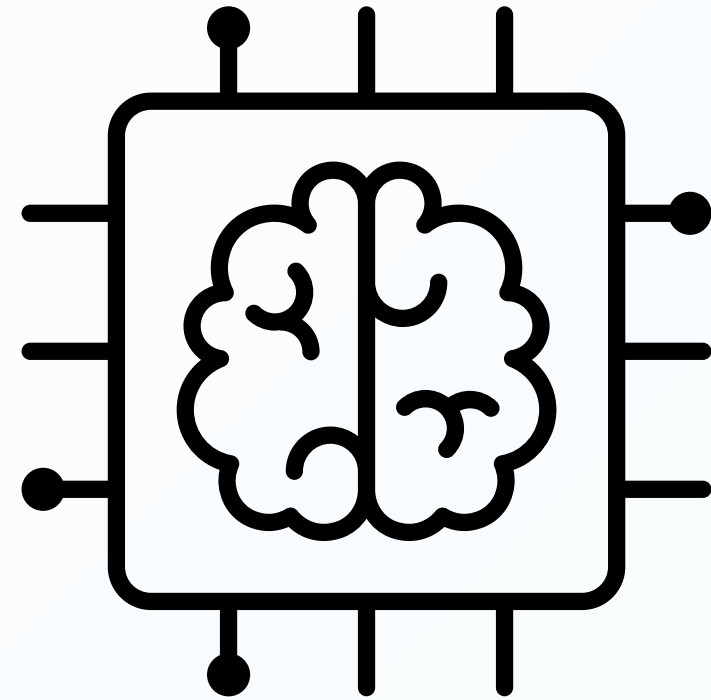
Calibrated Linear SVM

Margin-based text classifier with five-fold sigmoid calibration.

DistilBERT

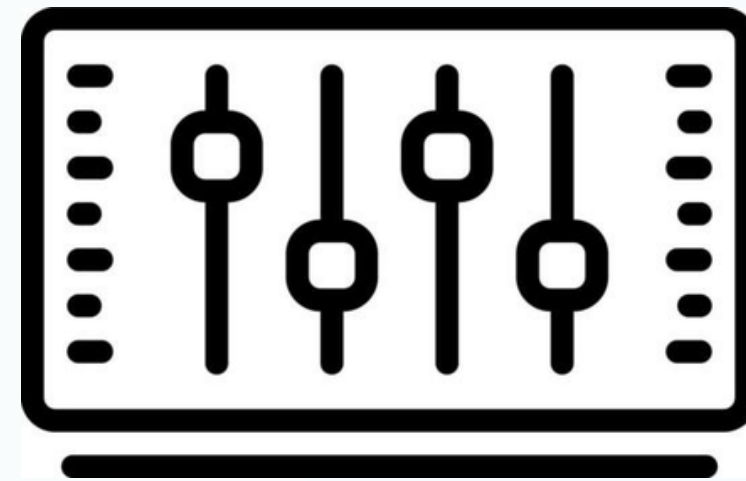
Fine-tuned transformer testing whether context improves class balance.

OUR SPLITTING CONFIG



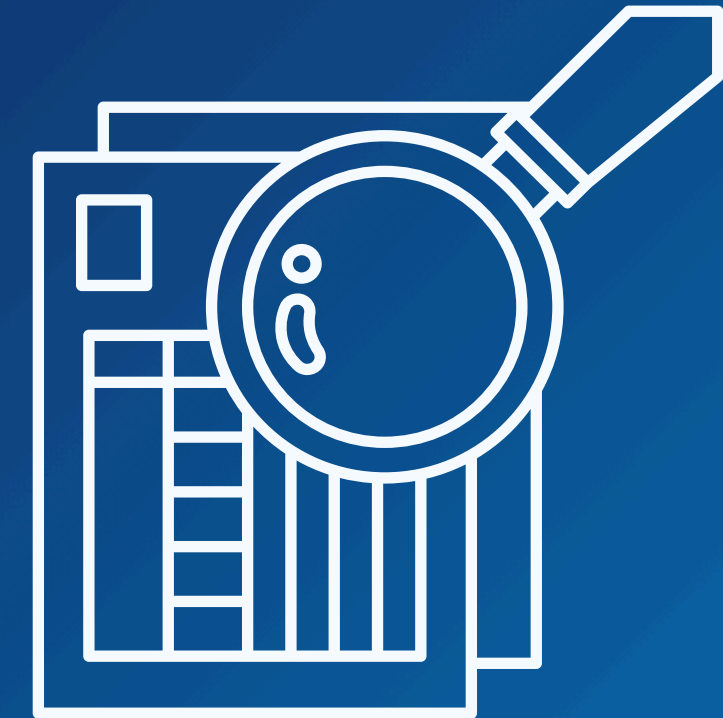
70%

Fit features, weights, parameters,
and calibration folds



15%

Select configurations by macro-F1



15%

450 held-out test
Opened once after freezing

CONFIGURATIONS WERE SELECTED ON VALIDATION MACRO-F1

MODEL	SELECTED CONFIGURATION
Naive Bayes	alpha = 0.1
Logistic Regression	C = 1.0 · balanced class weights · max_iter = 5000
Calibrated Linear SVM	C = 1.0 · balanced weights · five-fold sigmoid calibration
DistilBERT	LR 2e-5 · batch 8 · five epochs · 256 tokens · stride 64 · inverse-frequency weights

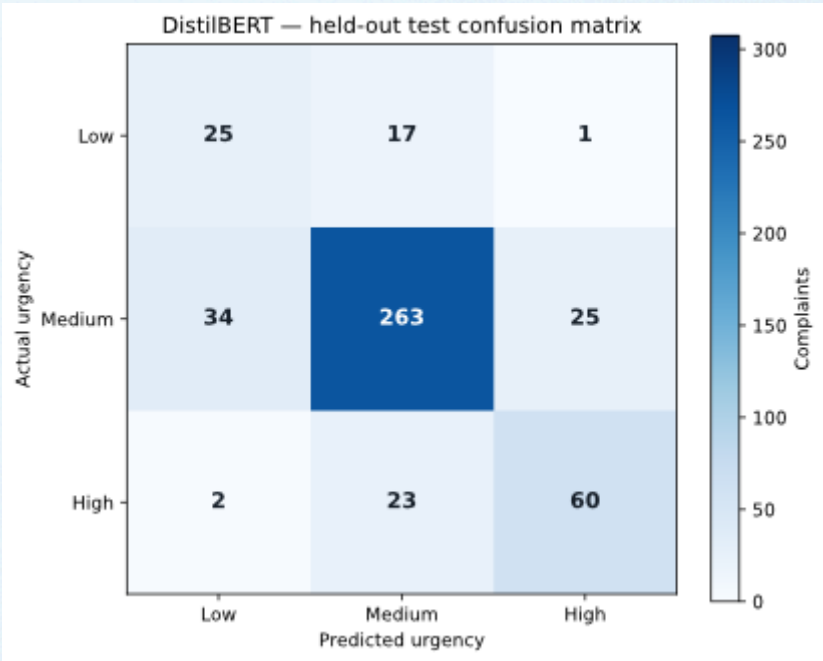
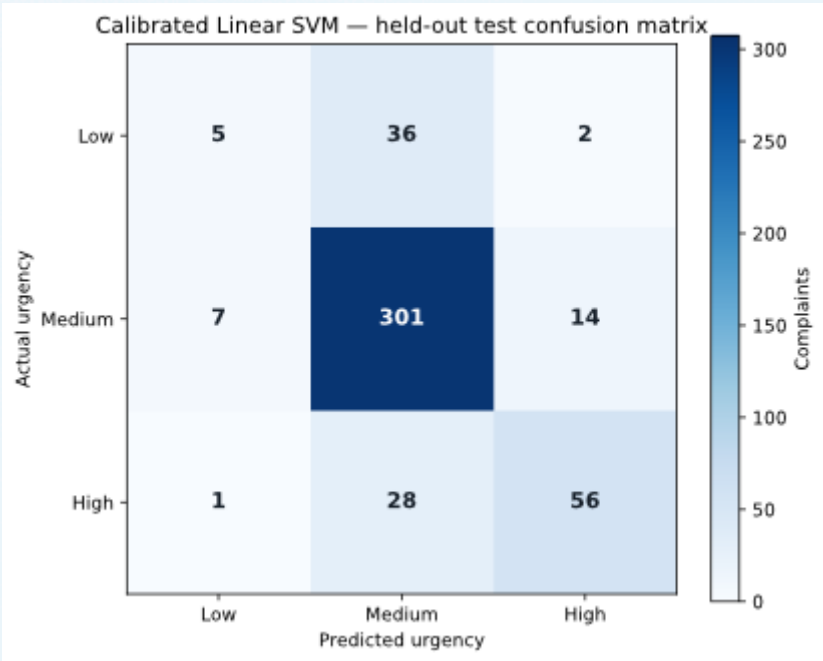
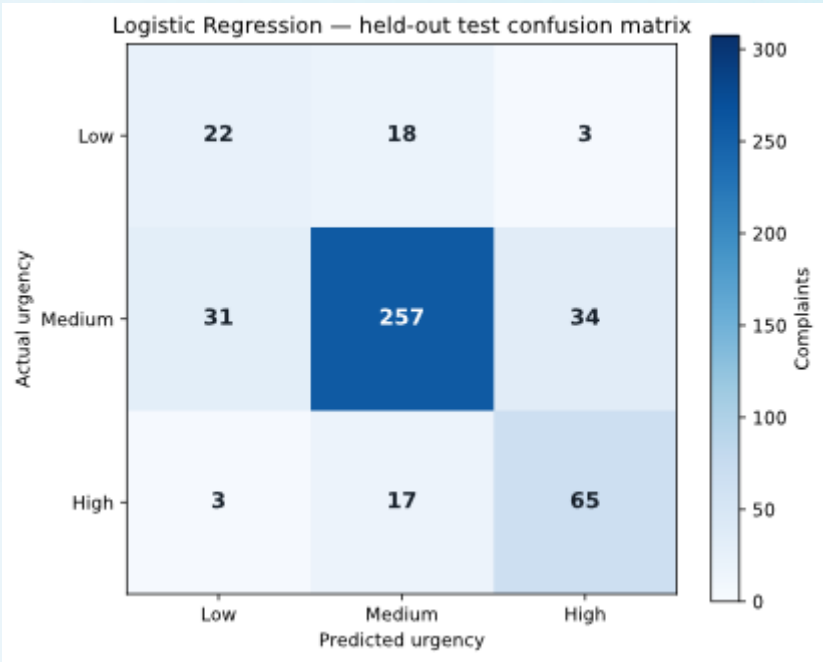
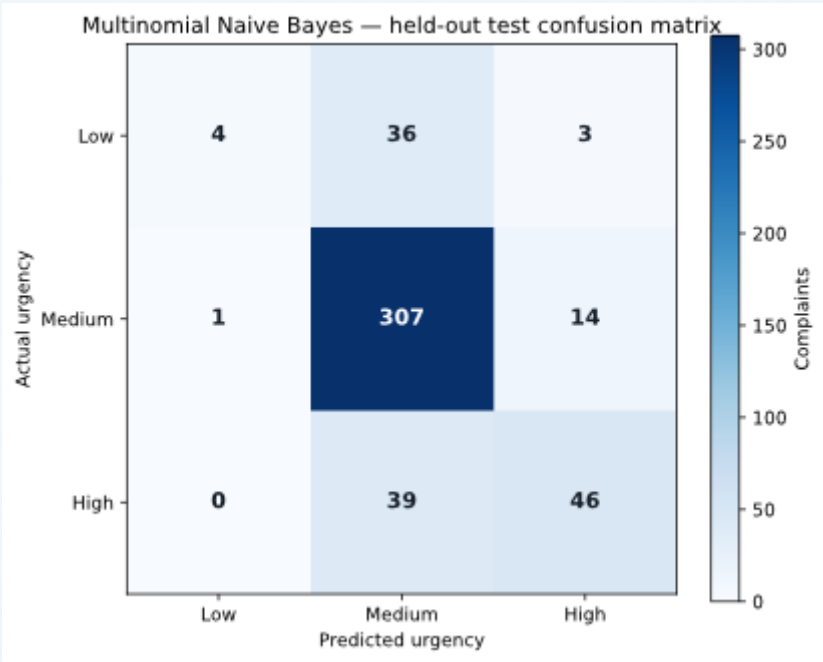
03

RESULTS & INTERPRETATION

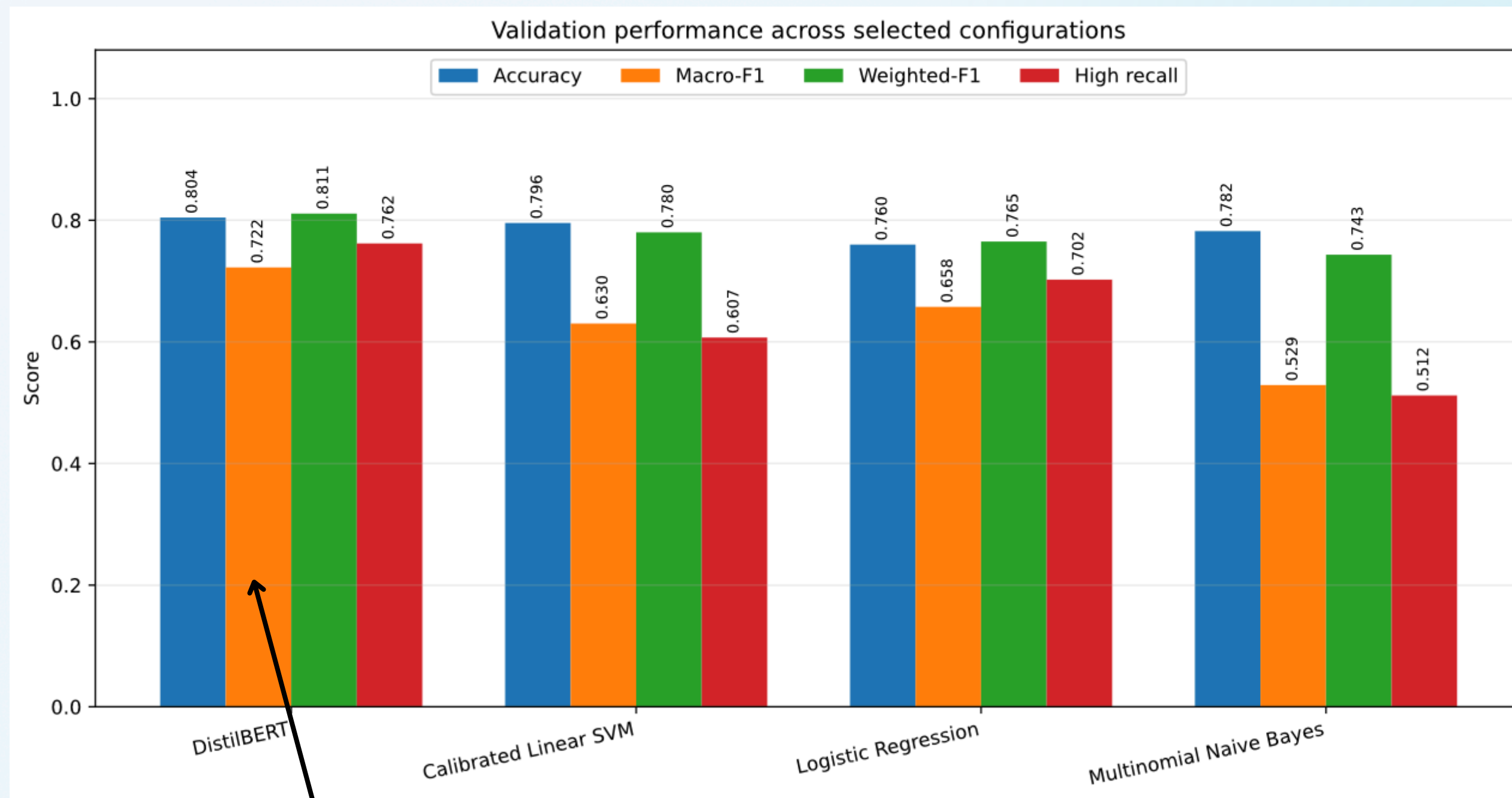
FULL HELD-OUT TEST COMPARISON

MODEL	ACCURACY	MACRO-F1	WEIGHTED-F1	HIGH RECALL	HIGH→LOW
Naive Bayes	0.7933	0.5535	0.7574	0.5412	0
Logistic Regression	0.7644	0.6589	0.7728	0.7647	3
Calibrated Linear SVM	0.8044	0.5894	0.7788	0.6588	1
DistilBERT	0.7733	0.6747	0.7807	0.7059	2

CONFUSION MATRICES

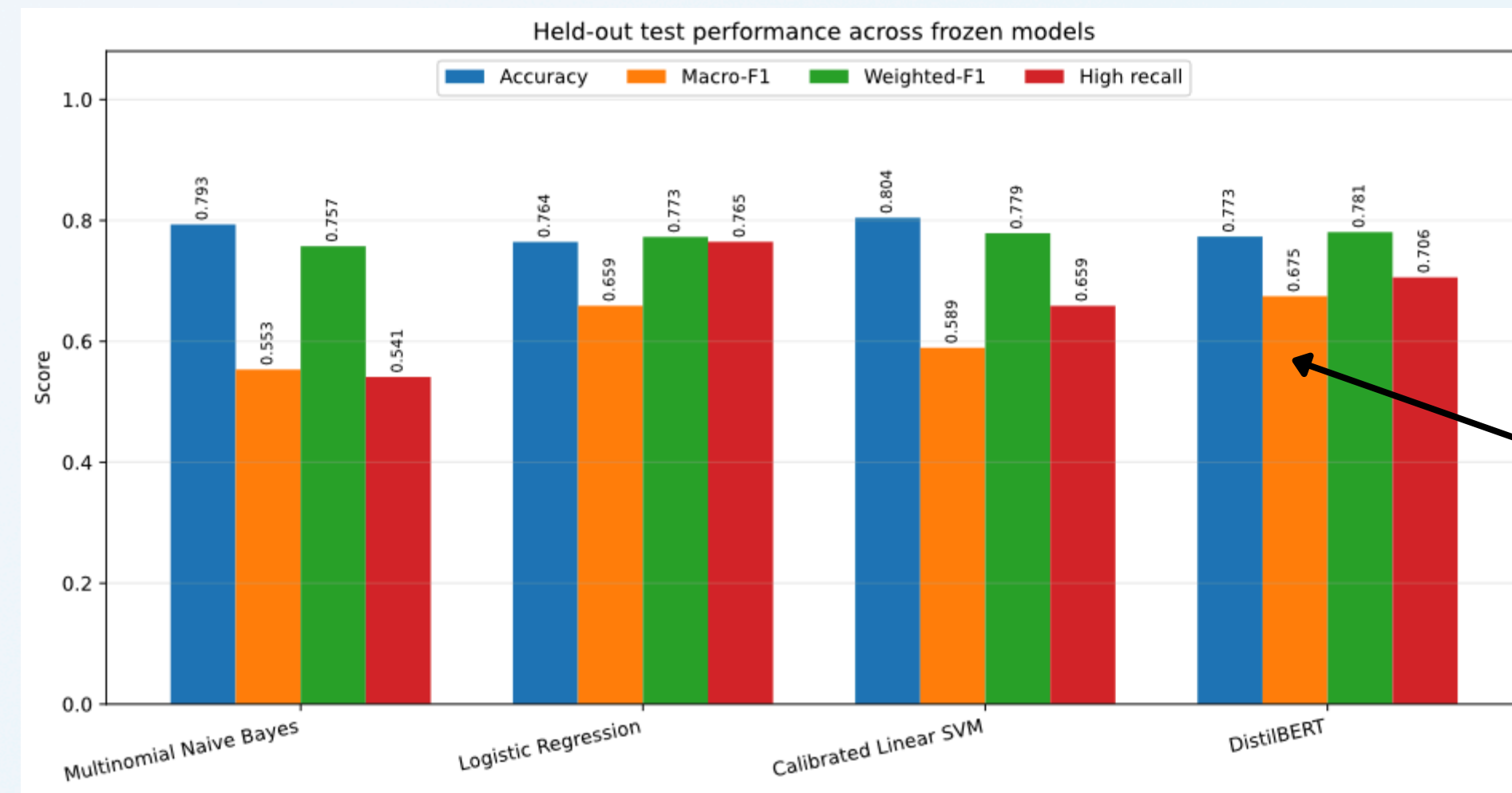


VALIDATION FAVORED DISTILBERT'S CLASS BALANCE



Validation macro-F1: **DistilBERT 0.7225** · Logistic Regression 0.6576 · SVM 0.6302 · Naive Bayes 0.5290

DISTILBERT ACHIEVED THE BEST BALANCED HELD-OUT PERFORMANCE



PRIMARY METRIC

0.6747

DistilBERT macro-F1
+0.0158 over Logistic Regression

Same 450 complaint IDs for every frozen model; no post-test tuning.

BUT THE HIGHEST-ACCURACY MODEL WAS NOT THE BEST MODEL

CALIBRATED SVM

0.8044

Highest test accuracy

BUT FOR LOW URGENCY

0.1786

Low-class F1·only 5 of 43 Low cases recovered

...Majority-class success can conceal operationally important failure.

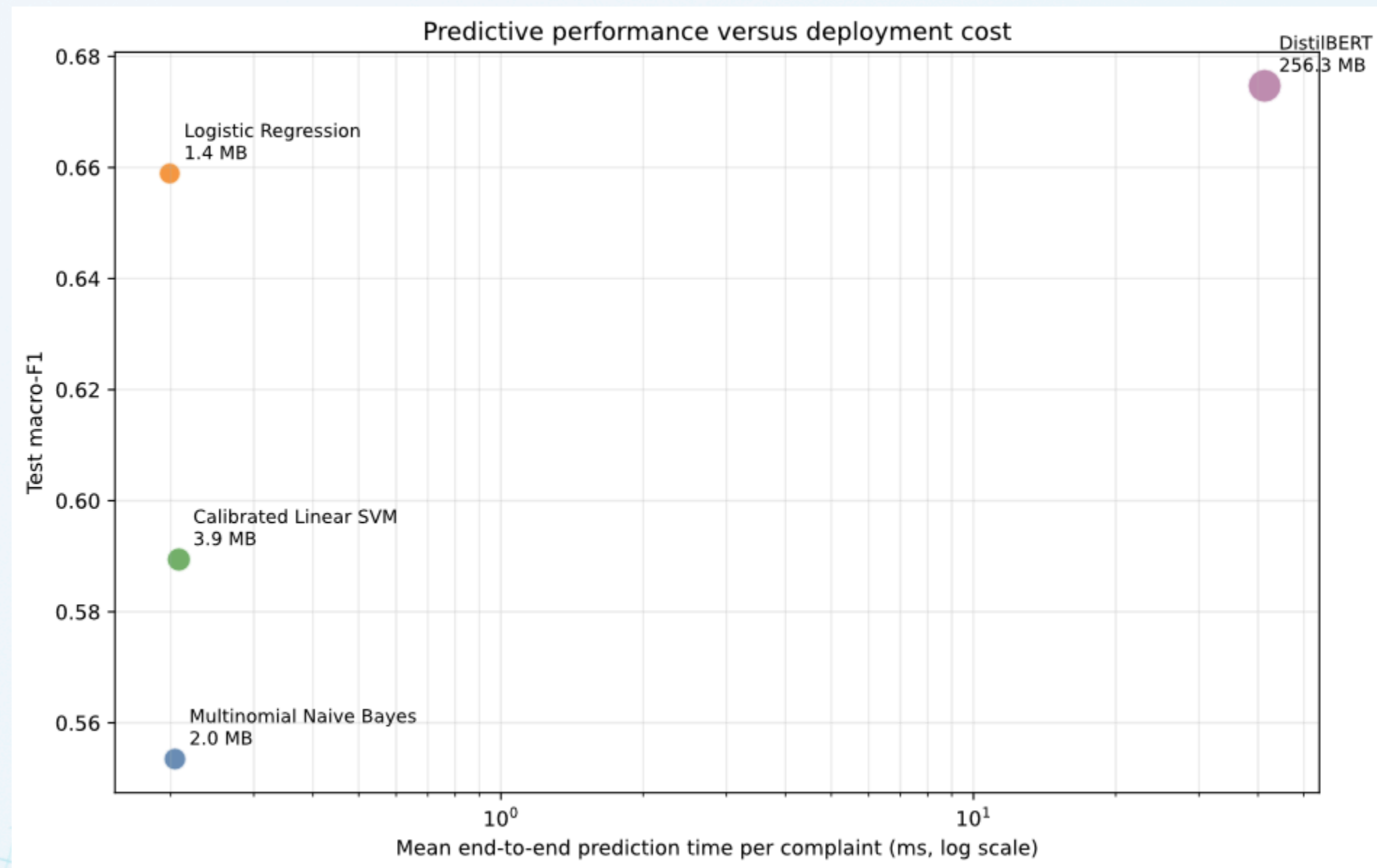
SAFETY METRICS CHANGED HOW THE MODELS WERE INTERPRETED



0.7647
Best High recall
Logistic Regression

Zero direct High→Low errors did not make Naive Bayes safe: 39 High cases were predicted Medium.

THE PERFORMANCE GAIN CARRIED A SUBSTANTIAL COMPUTATIONAL COST



207×

slower mean prediction than Logistic Regression

189×

larger persisted artifact set

ERRORS CLUSTERED AROUND IMPLICIT RISK, TIMING, AND LONG CONTEXT

● Actual High → predicted Low

Implicit identity misuse

An unauthorized inquiry was described as potential identity misuse, but routine credit-report language outweighed the urgency cue.

● Actual High → predicted Medium

Urgency diluted by background

A long mortgage narrative mixed history, allegations, and continuing risk, making the decisive passage harder to isolate.

Borderline and lengthy complaints remain strong candidates for mandatory human review.

04 PROTOTYPE SHOWCASE

THE PROTOTYPE BEGINS WITH AN OPERATIONAL COMPLAINT QUEUE

Complaint operations

Daily triage workspace

Queue

My cases

Escalations

Reports

Signed in as

Alex Peterson

Triage officer

Complaint queue

Review incoming complaints and act on the next priority.

+ New complaint

Search cases

Search by case ID, issue, or product

AI priority

All priorities

Status

All statuses

CASE AND ISSUE	AI PRIORITY	STATUS	ASSIGNEE	RECEIVED AND RESPONSE DUE	
CMP-2026-0184 Unauthorized card transaction	High	New	Alex Peterson	Received 22 Aug at 17:04 UTC Due 22 Aug at 17:04 UTC	→ Open
CMP-2026-0183 Mortgage payment history is incorrect	Medium	In review	Alex Peterson	Received 22 Aug at 16:04 UTC Due 25 Aug at 16:04 UTC	→ Open
CMP-2026-0182 Debt collection account is not recognized	High	Escalated	Unassigned	Received 22 Aug at 15:04 UTC Due 22 Aug at 15:04 UTC	→ Open
CMP-2026-0181 Request for credit report dispute information	Low	New	Alex Peterson	Received 22 Aug at 14:04 UTC Due 28 Aug at 14:04 UTC	→ Open
CMP-2026-0180 Overdraft fee appears incorrect	Medium	New	Unassigned	Received 22 Aug at 13:04 UTC Due 25 Aug at 13:04 UTC	→ Open
CMP-2026-0179 Account closed without explanation	High	Escalated	Alex Peterson	Received 22 Aug at 12:04 UTC Due 22 Aug at 12:04 UTC	→ Open
CMP-2026-0177 Routine statement and address request	Low	New	Unassigned	Received 22 Aug at 10:04 UTC Due 28 Aug at 10:04 UTC	→ Open

All visible case data are synthetic.

PREDICTIONS BECOME INPUTS TO A HUMAN-LED CASE WORKFLOW

Complaint operations
Daily triage workspace

- Queue
- My cases
- Escalations
- Reports

Signed in as
Alex Peterson
Triage officer

Complaint queue / Case workspace

← Previous case 1 of 7 → Next case

CMP-2026-0184
Unauthorized transactions | Debit card | Received 22 Aug at 16:44 UTC

Issue summary
Unauthorized card transaction
Source: Web form

Complaint narrative
An unauthorized card transaction remains on my account and the funds have not been returned.

Activity timeline
Complaint received
Received via Web form
22 Aug at 16:44 UTC

Add internal note
Internal note
Record a contact, decision, or next step.
Add note

AI priority
High
Priority score: 91%

Why this was flagged
Generate explanation
Weighted influencing terms
unauthorized (+0.82)
card (+0.66)

Recommended action
Escalate now and complete a same-day priority review.

Case details
Current assignee: Alex Peterson
Assign to
Assign a case owner

Response due: 23 Aug at 16:44 UTC
Status
New

Case actions
Mark in review
Escalate case

- AI priority and score
- Recommended action
- Assignee and status controls
- Internal notes and timeline
- On-demand “Why this was flagged” support

The officer remains responsible for the operational decision.

05 DECISION

THE RIGHT RECOMMENDATION DEPENDS ON THE OPERATING PRIORITY

PERFORMANCE RECOMMENDATION

DistilBERT

- Best macro-F1: 0.6747
- Strongest Low-class F1
- Selected for the academic prototype

LIGHTWEIGHT ALTERNATIVE

Logistic Regression

- Macro-F1: 0.6589
- Best High recall: 0.7647
- 1.36 MiB and ~207× faster prediction

DistilBERT's advantage is real but modest—not decisive under every deployment constraint.

STRONGER LABELS AND UNCERTAINTY HANDLING ARE THE NEXT PRIORITIES

CURRENT BOUNDARIES

- LLM-generated urgency labels
- One held-out test of 450 cases
- Only 43 Low and 85 High test examples
- Non-representative, unverified
- narratives Unmeasured calibration and fairness

FUTURE WORK

- Domain-expert annotation
- Confidence intervals and external validation
- Calibration, abstention, and thresholding
- Fairness and temporal-drift analysis
- Alternative long-document aggregation

06

CONCLUSION

Context matters

Urgency cannot be reduced to product or tone

DistillBERT leads

Best balanced held-out performance.

Humans decide

AI support triage; it does not determine outcomes.

THANK YOU